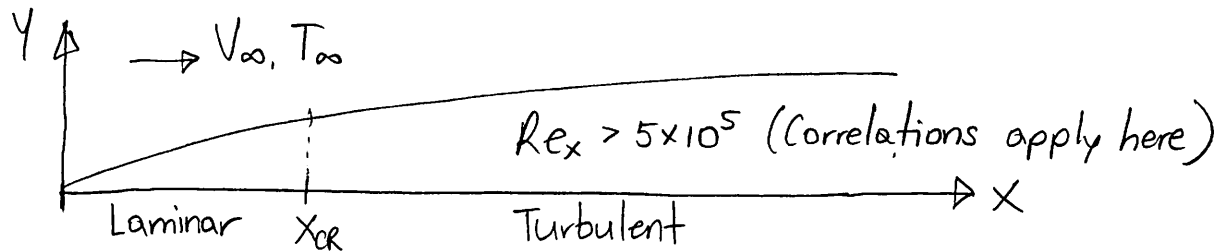


Note, so far, everything we've been covering has been in the turbulent region of the plate. But there is still a laminar section. How do we handle this?



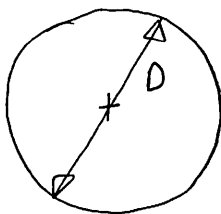
Mixed flow correlations exist (pg. 123 of notes)

$$\overline{C_{f,L}} = 0.074 Re_L^{-1/5} - 1742 Re_L^{-1} \quad Re_L < 10^8$$

$$\overline{Nu_L} = \frac{\bar{h} L}{k_f} = (0.037 Re_L^{4/5} - 871) Pr^{1/3} \quad \begin{matrix} Re_L < 10^8 \\ 0.6 < Pr < 60 \end{matrix}$$

Cylinders & Spheres (External flow)

V_{∞}, T_{∞}



$$\overline{Nu_0} = \frac{\bar{h} D}{k_f} = \frac{1}{0.8237 - \ln(Re_0 Pr)^{1/2}}$$

$\hookrightarrow$ Cylinder, $Re_0 Pr < 0.2$

$$\overline{Nu_0} = \frac{\bar{h} D}{k_f} = 0.3 + \frac{0.62 Re_0^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}}$$

$\hookrightarrow$ Cylinder, $Re_0 > 10^4$

$$\overline{Nu_0} = 2 + (0.4 Re_0^{1/2} + 0.06 Re_0^{2/3}) Pr^{0.4} \quad \begin{matrix} \Rightarrow \text{Sphere} \\ 3.5 < Re_0 < 8 \times 10^4 \\ 0.7 < Pr < 380 \end{matrix}$$

For rest of correlations, see pg. 123 of notes.

TABLE 8.4 Summary of convection correlations for flow in a circular tube^a

Correlation		Conditions
$f = 64/Re_D$	(8.19)	Laminar, fully developed
$Nu_D = 4.36$	(8.53)	Laminar, fully developed, uniform q_s''
$Nu_D = 3.66$	(8.55)	Laminar, fully developed, uniform T_s
$\bar{Nu}_D = 3.66 + \frac{0.0668 Gz_D}{1 + 0.04 Gz_D^{-2/3}}$	(8.57)	Laminar, thermal entry (or combined entry with $Pr \geq 5$), uniform T_s , $Gz_D = (D/x) Re_D Pr$
$\bar{Nu}_D = \frac{\frac{3.66}{\tanh[2.264 Gz_D^{-1/3} + 1.7 Gz_D^{-2/3}]} + 0.0499 Gz_D \tanh(Gz_D^{-1})}{\tanh(2.432 Pr^{1/6} Gz_D^{-1/6})}$	(8.58)	Laminar, combined entry, $Pr \geq 0.1$, uniform T_s , $Gz_D = (D/x) Re_D Pr$
$\frac{1}{\sqrt{f}} = -2.0 \log \left[\frac{e/D}{3.7} + \frac{2.51}{Re_D \sqrt{f}} \right]$	(8.20) ^c	Turbulent, fully developed
$f = (0.790 \ln Re_D - 1.64)^{-2}$	(8.21) ^c	Turbulent, fully developed, smooth walls, $3000 \leq Re_D \leq 5 \times 10^6$
$Nu_D = 0.023 Re_D^{4/5} Pr^n$	(8.60) ^d	Turbulent, fully developed, $0.6 \leq Pr \leq 160$, $Re_D \geq 10,000$, $(L/D) \geq 10$, $n = 0.4$ for $T_s > T_m$ and $n = 0.3$ for $T_s < T_m$
$Nu_D = 0.027 Re_D^{4/5} Pr^{1/3} \left(\frac{\mu}{\mu_s} \right)^{0.14}$	(8.61) ^d	Turbulent, fully developed, $0.7 \leq Pr \leq 16,700$, $Re_D \geq 10,000$, $L/D \geq 10$
$Nu_D = \frac{(f/8)(Re_D - 1000) Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}$	(8.62) ^d	Turbulent, fully developed, $0.5 \leq Pr \leq 2000$, $3000 \leq Re_D \leq 5 \times 10^6$, $(L/D) \geq 10$
$Nu_D = 4.82 + 0.0185(Re_D Pr)^{0.827}$	(8.64)	Liquid metals, turbulent, fully developed, uniform q_s'' , $3.6 \times 10^3 \leq Re_D \leq 9.05 \times 10^5$, $3 \times 10^{-3} \leq Pr \leq 5 \times 10^{-2}$, $10^2 \leq Re_D Pr \leq 10^4$
$Nu_D = 5.0 + 0.025(Re_D Pr)^{0.8}$	(8.65)	Liquid metals, turbulent, fully developed, uniform T_s , $Re_D Pr \geq 100$

^aThe mass transfer correlations may be obtained by replacing Nu_D and Pr by Sh_D and Sc , respectively.

^bProperties in Equations 8.53, 8.55, 8.60, 8.61, 8.62, 8.64, and 8.65 are based on T_m ; properties in Equations 8.19, 8.20, and 8.21 are based on $T_f = (T_s + T_m)/2$; properties in Equations 8.57 and 8.58 are based on $\bar{T}_m = (T_{m,i} + T_{m,o})/2$.

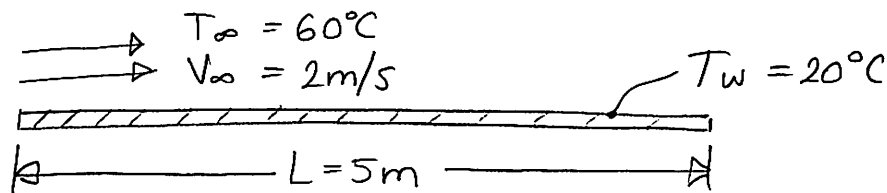
^cEquation 8.20 pertains to smooth or rough tubes. Equation 8.21 pertains to smooth tubes.

^dAs a first approximation, Equations 8.60, 8.61, or 8.62 may be used to evaluate the average Nusselt number $\bar{Nu}_D$ over the entire tube length, if $(L/D) \geq 10$. The properties should then be evaluated at the average of the mean temperature, $\bar{T}_m = (T_{m,i} + T_{m,o})/2$.

^eFor tubes of noncircular cross section, $Re_D = D_h \mu_m / \nu$, $D_h = 4A_c/P$, and $\mu_m = m/\rho A_c$. Results for fully developed laminar flow are provided in Table 8.1. For turbulent flow, Equation 8.60 may be used as a first approximation.

NOTE* For a summary of external flow correlations, go to page # 123 of the notes (Table 7.7 or 7.9)

Example | Engine oil at 60°C flows over a 5m long flat plate (20°C) with velocity of 2m/s. Determine total drag force and heat transfer per unit width.



$$T_f = \bar{T} = \frac{T_w + T_\infty}{2} = 40^\circ\text{C}$$

$$\rho_{\text{oil}} = 876 \text{ kg/m}^3$$

$$k_{\text{oil}} = 0.144 \text{ W/m}\cdot\text{K}$$

$$Pr = 2870$$

$$\nu_{\text{oil}} = 242 \times 10^{-6} \text{ m}^2/\text{s}$$

Let's check our flow regime:

$$Re_L = \frac{V_\infty L}{\nu_{\text{oil}}} = 4.13 \times 10^4 < 5 \times 10^5 \text{ (Laminar)}$$

We know from pg. 123 of notes:

$$\bar{C}_f = \frac{\bar{C}_0}{\frac{1}{2} \rho V_\infty^2} = 1.328 Re_L^{-0.5} = 0.00653$$

$$F_D = \bar{C}_f \cdot A \cdot \frac{\rho V_\infty^2}{2} = (0.00653)(5\text{m} \cdot 1\text{m}) \frac{(876 \text{ kg/m}^3)(2\text{m/s})^2}{2}$$

$$\boxed{F_D = 57.2 \text{ N}}$$

For heat transfer:

$$\overline{Nu}_L = \frac{\bar{h} L}{k_{oil}} = 0.664 Re_L^{1/2} Pr^{1/3} = 1918$$

$$\bar{h} = 55.2 \text{ W/m}^2 \cdot \text{K}$$

$$Q = \bar{h} A (T_\infty - T_w) = 11.04 \text{ kW}$$

Now what if we increased the flow rate by 15x?
Do we expect friction & heat transfer to increase by 15x?

$$Re_L = \frac{V_\infty L}{\nu_{oil}} = 6.2 \times 10^5 > Re_{cr} \quad (\text{Turbulent flow})$$

$$\bar{C}_f = 0.074 Re_L^{-1/5} - 1742 Re_L^{-1} \quad (Re_L < 10^8) \Rightarrow \text{Mixed flow}$$

$$\bar{C}_f = 0.00233$$

$$F_D = \bar{C}_f \cdot A \cdot \frac{\rho V_\infty^2}{2} = (0.00233)(5 \text{ m}^2) \frac{(876 \text{ kg/m}^3)(30 \text{ m/s})^2}{2}$$

$$F_D = 4.59 \text{ kN}$$

For heat transfer:

$$\overline{Nu}_L = \frac{\bar{h} L}{k_{oil}} = (0.037 Re_L^{4/5} - 871) Pr^{1/3} = 10255.2$$

$$\bar{h} = 295.3 \text{ W/m}^2 \cdot \text{K}$$

$$Q = \bar{h} A (T_\infty - T_w) = 59.1 \text{ kW}$$

$$\frac{F_{D, \text{TURB}}}{F_{D, \text{LAM}}} = 80.2$$

$$\frac{Q_{\text{TURB}}}{Q_{\text{LAM}}} = 5.4$$